



Tangent Lines to Curves Using Derivatives

Calculus Worksheet · Grade 11-12

Name: _____

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Score: / 10

Learning Objectives

- Apply derivative rules to trigonometric functions
- Evaluate derivatives at specific points to determine slope
- Write equations of tangent lines using point-slope form

Find the equation of the tangent line to each curve at the given point, expressing your answer in point-slope form.

1. Find the equation of the tangent line to the curve at the given point.

$$y = \sec x \text{ at } \left(\frac{\pi}{3}, 2\right)$$

Answer: _____

2. Find the equation of the tangent line to the curve at the given point.

$$y = \sec x - 2\cos x \text{ at } \left(\frac{\pi}{3}, 1\right)$$

Answer: _____

3. Find the equation of the tangent line to the curve at the given point.

$$y = \sin x \text{ at } \left(\frac{\pi}{6}, \frac{1}{2}\right)$$

Answer: _____

4. Find the equation of the tangent line to the curve at the given point.

$$y = \cos x \text{ at } \left(\frac{\pi}{4}, \frac{\sqrt{2}}{2}\right)$$

Answer: _____

5. Find the equation of the tangent line to the curve at the given point.

$$y = \tan x \text{ at } \left(\frac{\pi}{4}, 1\right)$$

Answer: _____

6. Find the equation of the tangent line to the curve at the given point.

$$y = \csc x \text{ at } \left(\frac{\pi}{6}, 2\right)$$

Answer: _____



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7. Find the equation of the tangent line to the curve at the given point.

$$y = x^2 + 3x \text{ at } (1, 4)$$

Answer: _____

8. Find the equation of the tangent line to the curve at the given point.

$$y = \sqrt{x} \text{ at } (4, 2)$$

Answer: _____

9. Find the equation of the tangent line to the curve at the given point.

$$y = \ln x \text{ at } (1, 0)$$

Answer: _____

10. Find the equation of the tangent line to the curve at the given point.

$$y = e^x \text{ at } (0, 1)$$

Answer: _____





Students should be fluent with unit-circle values and derivative rules for trigonometric functions before attempting these problems.

Solutions

1. Find the equation of the tangent line to the curve at the given point.

$$y = \sec x \text{ at } \left(\frac{\pi}{3}, 2\right)$$

→ Differentiate y with respect to x to get y prime equal to secant x times tangent x .

→ Evaluate the derivative at x equal to π over three: secant of π over three is 2 and tangent of π over three is square root of three.

→ Multiply to find the slope m equals 2 times square root of three.

→ Substitute the point and slope into the point-slope form y minus y one equals m times x minus x one.

Answer: $y - 2 = 2\sqrt{3}\left(x - \frac{\pi}{3}\right)$

2. Find the equation of the tangent line to the curve at the given point.

$$y = \sec x - 2\cos x \text{ at } \left(\frac{\pi}{3}, 1\right)$$

→ Differentiate term by term: derivative of secant x is secant x tangent x and derivative of negative two cosine x is two sine x .

→ Combine to get y prime equal to secant x tangent x plus two sine x .

→ Evaluate at π over three: secant π over three is 2, tangent π over three is square root of three, and sine π over three is square root of three over two.

→ Compute the slope: 2 times square root of three plus two times square root of three over two equals three square root of three.

→ Use point-slope form with the given point and slope to write the tangent line.

Answer: $y - 1 = (2\sqrt{3} + \sqrt{3})\left(x - \frac{\pi}{3}\right)$

3. Find the equation of the tangent line to the curve at the given point.

$$y = \sin x \text{ at } \left(\frac{\pi}{6}, \frac{1}{2}\right)$$

→ Compute the derivative: y prime equals cosine x .

→ Evaluate cosine at π over six to find the slope, which is square root of three over two.

→ Plug the slope and point into the point-slope formula.

Answer: $y - \frac{1}{2} = \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right)$

4. Find the equation of the tangent line to the curve at the given point.

$$y = \cos x \text{ at } \left(\frac{\pi}{4}, \frac{\sqrt{2}}{2}\right)$$

→ Differentiate to obtain y prime equal to negative sine x .

→ Evaluate at π over four: negative sine of π over four equals negative square root of two over two.

→ Substitute the slope and the point into the point-slope form.

Answer: $y - \frac{\sqrt{2}}{2} = -\frac{\sqrt{2}}{2}\left(x - \frac{\pi}{4}\right)$



5. Find the equation of the tangent line to the curve at the given point.

$$y = \tan x \text{ at } \left(\frac{\pi}{4}, 1\right)$$

→ Use the derivative rule for tangent: y prime equals secant squared x .

→ Evaluate at π over four: secant of π over four is square root of two, so secant squared is 2.

→ Apply the slope of 2 and the given point to the point-slope formula.

Answer: $y - 1 = 2\left(x - \frac{\pi}{4}\right)$

6. Find the equation of the tangent line to the curve at the given point.

$$y = \csc x \text{ at } \left(\frac{\pi}{6}, 2\right)$$

→ Differentiate: y prime equals negative cosecant x times cotangent x .

→ Evaluate at π over six: cosecant of π over six is 2 and cotangent of π over six is square root of three.

→ Multiply with the negative sign to get slope equal to negative two square root of three.

→ Write the tangent line using point-slope form.

Answer: $y - 2 = -2\sqrt{3}\left(x - \frac{\pi}{6}\right)$

7. Find the equation of the tangent line to the curve at the given point.

$$y = x^2 + 3x \text{ at } (1, 4)$$

→ Apply the power rule to find y prime equal to two x plus three.

→ Evaluate at x equal to one to find the slope equal to five.

→ Insert the slope and the point into the point-slope formula.

Answer: $y - 4 = 5(x - 1)$

8. Find the equation of the tangent line to the curve at the given point.

$$y = \sqrt{x} \text{ at } (4, 2)$$

→ Rewrite y as x to the one-half and differentiate to get y prime equal to one over two times the square root of x .

→ Evaluate at x equal to four: one over two times two equals one fourth.

→ Use the slope of one fourth with the given point in point-slope form.

Answer: $y - 2 = \frac{1}{4}(x - 4)$

9. Find the equation of the tangent line to the curve at the given point.

$$y = \ln x \text{ at } (1, 0)$$

→ Recall the derivative of natural log of x is one over x .

→ Evaluate at x equal to one to find the slope equal to one.

→ Substitute into point-slope form to write the equation.

Answer: $y - 0 = 1(x - 1)$

10. Find the equation of the tangent line to the curve at the given point.

$$y = e^x \text{ at } (0, 1)$$

→ The derivative of e to the x is e to the x itself.

→ Evaluate at x equal to zero: e to the zero equals one, so the slope is one.

→ Apply the slope and point to the point-slope form.

Answer: $y - 1 = 1(x - 0)$

